AI-900: Microsoft Azure AI Fundamentals (Exam Study Source Material)

Last Updated: May 2025
Exam Version: GenAI Enhanced Edition

EXECUTIVE SUMMARY

This document serves as the primary study material for the Microsoft AI-900 exam. The exam measures the candidate's ability to describe Artificial Intelligence workloads and considerations, fundamental principles of machine learning on Azure, computer vision, natural language processing (NLP), and Generative AI.

Critical Note for 2025: The exam has been significantly updated as of May 2, 2025. The most notable change is the increased weight of Generative AI content (now 20-25% of the exam) and the introduction of "Azure AI Foundry" which replaces previous unified studio concepts.

SECTION 1: CRITICAL UPDATES (MAY 2025)

1. Azure AI Foundry

- **Definition:** The new unified platform for building, evaluating, and deploying AI models.
- **Context:** Wherever older study guides mention "Azure AI Studio" or generic management portals, the correct answer is now likely "Azure AI Foundry."
- **Key Feature:** The **Model Catalog** within Foundry is the central repository for discovering foundation models (like GPT-4, Llama 2, etc.).

2. Generative AI Weight Increase

- **Change:** Generative AI is now the single largest domain (20-25%).
- **Implication:** Candidates must prioritize understanding LLMs (Large Language Models), prompt engineering, and the Copilot ecosystem over deeper technical details of traditional regression models.

SECTION 2: AI WORKLOADS & RESPONSIBLE AI (15-20%)

Core AI Workloads

1. **Machine Learning:** The foundation of AI. It involves using data to train models to make predictions (e.g., predicting house prices).
2. **Computer Vision:** Software that can "see" by interpreting images and video (e.g.,

unlocking a phone with a face).
3. **Natural Language Processing (NLP):** Software that can "read" and "speak" (e.g., translation apps, subtitles).
4. **Document Intelligence:** A specialized workload for extracting data from forms, receipts, and invoices.
5. **Generative AI:** Systems that create *new* content (text, images, code) rather than just analyzing existing data.

The 6 Principles of Responsible AI (Memorize Required)

Microsoft defines six principles that must guide AI development to prevent harm:

1. **Fairness:** AI must treat everyone fairly and avoid bias (e.g., a loan approval model shouldn't discriminate based on gender).
2. **Reliability & Safety:** AI must perform reliably and safely, even in unexpected conditions (e.g., an autonomous car must not crash if it rains).
3. **Privacy & Security:** AI must respect user privacy and data must be secure (e.g., medical data must be protected).
4. **Inclusiveness:** AI should empower everyone, including those with disabilities (e.g., providing screen readers or accessible interfaces).
5. **Transparency:** AI systems should be understandable. Users should know how a decision was made (e.g., explaining why a bank loan was denied).
6. **Accountability:** Humans must remain accountable for AI systems. Designers and developers are responsible for the AI's actions.

Official Study Module: [Introduction to AI concepts]

SECTION 3: MACHINE LEARNING FUNDAMENTALS (15-20%)

3 Main Types of Machine Learning

1. **Regression**
   - **Goal:** Predict a numeric value (a number).
   - **Keywords:** "How much?", "How many?", "Temperature", "Price", "Sales Forecast".
   - *Example:* Predicting the price of a used car based on its mileage.
2. **Classification**
   - **Goal:** Predict a category or label.
   - **Binary Classification:** Only two options (Yes/No, Spam/Not Spam).
   - **Multiclass Classification:** More than two options (Red/Blue/Green, Cat/Dog/Bird).
   - *Example:* Is this credit card transaction fraudulent? (Yes/No).
3. **Clustering**
   - **Goal:** Group similar items together based on features. Unlike Regression/Classification, you don't know the answer beforehand (Unsupervised Learning).

    - **Keywords:** "Segmentation", "Grouping", "Separating".
    - *Example:* Grouping customers into different marketing segments based on purchasing habits.

Azure Machine Learning Tools

- **Automated Machine Learning (AutoML):** A feature in Azure ML that automates the time-consuming tasks of selecting algorithms and tuning hyperparameters. Best for users who want quick results with less code.
- **Azure Machine Learning Designer:** A drag-and-drop visual interface for building models.

Official Study Module: Introduction to machine learning concepts

SECTION 4: COMPUTER VISION (15-20%)

Core Tasks & Services

- **Image Classification:** The machine looks at an image and says "That is a dog." (Broad label).
    - *Service:* Azure AI Vision.
- **Object Detection:** The machine looks at an image and draws a box around the dog and says "Dog is here" and "Cat is here." (Location + Label).
    - *Service:* Azure AI Vision.
- **Optical Character Recognition (OCR):** The machine extracts text from an image (e.g., reading a street sign or a scanned PDF).
    - *Service:* Azure AI Vision (Read API).
- **Facial Detection vs. Analysis:**
    - *Detection:* Finding a face in an image (Service: Azure AI Vision).
    - *Analysis:* Determining emotion, age, or identity (Service: Azure AI Face). Note: Face analysis is heavily restricted due to Responsible AI principles.

Official Study Module: Get started with computer vision in Azure

SECTION 5: NATURAL LANGUAGE PROCESSING (NLP) (15-20%)

Core Tasks & Services

- **Sentiment Analysis:** Determining if text is Positive, Negative, or Neutral.
    - *Service:* Azure AI Language.
- **Key Phrase Extraction:** Pulling out the main topics from a paragraph.
    - *Service:* Azure AI Language.
- **Named Entity Recognition (NER):** Identifying specific entities like People, Places, Dates, or Organizations in text.

    - *Service:* Azure AI Language.
- **Speech-to-Text & Text-to-Speech:** Converting spoken audio to written text and vice versa.
    - *Service:* Azure AI Speech.
- **Translation:** Converting text from one language to another.
    - *Service:* Azure AI Translator.

Official Study Module: [Get started with natural language processing in Azure]

SECTION 6: GENERATIVE AI (20-25%) - HIGH PRIORITY

Key Concepts

- **Large Language Models (LLMs):** AI models trained on vast amounts of data to understand and generate human-like language (e.g., GPT-4).
- **Tokenization:** The process of breaking text into smaller pieces (tokens) for the model to process. One token is roughly 4 characters or 0.75 words.
- **Prompt Engineering:** The skill of designing inputs (prompts) to guide the model to produce the best possible output.
    - *Grounding:* Adding specific data to a prompt so the model answers based on facts, not hallucinations.

Azure Ecosystem for GenAI

- **Azure OpenAI Service:** Provides REST API access to OpenAI's powerful language models (GPT-4, DALL-E) with the security and compliance of Azure.
- **Copilot:** An AI application that uses GenAI to assist users with complex tasks (e.g., GitHub Copilot helps write code; Microsoft 365 Copilot helps write documents).
- **Azure AI Foundry Model Catalog:** The place to explore, evaluate, and deploy foundation models.

Responsible Generative AI

GenAI introduces specific risks that must be managed:

- **Hallucinations:** The model making up false information confidently.
- **Harmful Content:** Generating violent or hateful text.
- **Mitigation:** Azure AI Content Safety is a service used to filter inputs and outputs for harmful content.

Official Study Module: [Introduction to generative AI and agents]

APPENDIX: QUICK EXAM CHEAT SHEET

| If the scenario is... | The Answer is likely... |

Predicting a number (price, temp)	Regression
Predicting a category (Yes/No)	Classification
Grouping items without labels	Clustering
Reading text from a photo	OCR / Azure AI Vision
Analyzing emotion in a face	Azure AI Face
Translating spoken audio	Azure AI Speech
Building a chat bot	Azure AI Language / Azure OpenAI
Filtering harmful content	Azure AI Content Safety
Unified interface for AI models	Azure AI Foundry
Automating algorithm selection	Automated ML (AutoML)

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